Date & Time Functions in Excel

Module 6: Working Days & Business Deadline Functions

In the previous module, we covered:

- DAYS
- DATEDIF

These functions calculate differences between dates, but they count **all calendar days**, including Saturdays and Sundays.

In real businesses, this is often not enough.

For example:

A ticket was opened on Friday and completed on Monday. Did it take 3 calendar days or 1 working day?

The answer depends on business rules.

This is where the following functions become important:

- NETWORKDAYS
- NETWORKDAYS.INTL
- WORKDAY
- WORKDAY.INTL

These functions are widely used in:

- Project management
- SLA tracking
- Employee attendance
- Payroll processing
- Delivery planning
- Financial reporting
- Ticket resolution analysis
- Business deadline calculations

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17. NETWORKDAYS Function

The NETWORKDAYS function calculates the number of **working days between two dates**, automatically excluding Saturdays and Sundays.

It can also optionally exclude specified holidays.

Purpose

Used to calculate actual business days between a start date and an end date.

Syntax

=NETWORKDAYS(start_date,end_date,[holidays])

Syntax Explanation

Argument	Meaning
start_date	Date when work begins
end_date	Date when work ends
[holidays]	Optional range containing holiday dates to exclude

The [holidays] argument is optional.

Basic Example

Suppose:
Start Date = 06-Jul-2026 (Monday)
End Date = 10-Jul-2026 (Friday)
Formula:
=NETWORKDAYS(DATE(2026,7,6),DATE(2026,7,10))
Result:
5
All five days are working days.

Example Including a Weekend

Suppose:
Start Date = 06-Jul-2026 (Monday)
End Date = 13-Jul-2026 (Monday)
There are 8 calendar days when counting both start and end dates, but Saturday and Sunday are excluded.
Formula:
=NETWORKDAYS(DATE(2026,7,6),DATE(2026,7,13))
Result:
6

Example Using Our Business Dataset

In our dataset:

- D2 = Start Date
- F2 = End Date

Formula:
=NETWORKDAYS(D2,F2)
This calculates the number of working days from ticket creation to ticket completion.

Business Example: Ticket Resolution in Working Days

Suppose:
Start Date = 03-Jul-2026 (Friday)
End Date = 06-Jul-2026 (Monday)
Formula:
=NETWORKDAYS(D2,F2)
Result:
2
Because both Friday and Monday are counted as working days.

Important Interview Point

NETWORKDAYS includes both the start date and end date if they are working days.
This is important because many students expect the answer to be 1, based on simple date subtraction.

NETWORKDAYS with Holidays

Suppose the company maintains a holiday list in:
L2:L10
Formula:
=NETWORKDAYS(D2,F2,L2:L10)
Now Excel excludes:

- Saturdays
- Sundays
- Dates listed in L2:L10

Business Example

Suppose:
Start Date = 06-Jul-2026
End Date = 10-Jul-2026
And:
08-Jul-2026 = Company Holiday
Without holiday exclusion:
=NETWORKDAYS(D2,F2)
Result:
5
With holiday exclusion:
=NETWORKDAYS(D2,F2,L2:L10)
Result:
4

Common Business Uses of NETWORKDAYS

- Ticket resolution time
- Employee working days
- Project duration
- SLA compliance
- Vendor turnaround time
- Delivery lead time
- Payroll calculations

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18. NETWORKDAYS.INTL Function

The NETWORKDAYS.INTL function calculates working days between two dates while allowing you to define **custom weekend patterns**.
Unlike NETWORKDAYS, it does not assume that weekends are always Saturday and Sunday.

Purpose

Used for organizations that follow alternative working schedules.
Examples:

- Friday–Saturday weekend
- Sunday-only weekend
- Monday-only weekend
- Six-day working week
- Custom non-working days

Syntax

=NETWORKDAYS.INTL(start_date,end_date,[weekend],[holidays])

Syntax Explanation

Argument	Meaning
start_date	Beginning date
end_date	Ending date
[weekend]	Defines which days are weekends
[holidays]	Optional holiday range

Common Weekend Codes

Weekend Code	Weekend Days
1	Saturday and Sunday
2	Sunday and Monday
3	Monday and Tuesday
4	Tuesday and Wednesday
5	Wednesday and Thursday
6	Thursday and Friday
7	Friday and Saturday
11	Sunday only
12	Monday only
13	Tuesday only
14	Wednesday only
15	Thursday only
16	Friday only
17	Saturday only

Example: Friday and Saturday as Weekend

=NETWORKDAYS.INTL(D2,F2,7)

Here:

- D2 = Start Date
- F2 = End Date
- 7 = Friday and Saturday are weekends

Example: Sunday-Only Weekend

=NETWORKDAYS.INTL(D2,F2,11)

Here, Monday through Saturday are treated as working days.

Using Holidays

=NETWORKDAYS.INTL(D2,F2,1,L2:L10)

This:

- Uses Saturday and Sunday as weekends.
- Excludes holidays stored in L2:L10.

NETWORKDAYS vs NETWORKDAYS.INTL

NETWORKDAYS	NETWORKDAYS.INTL
Assumes Saturday and Sunday weekend	Supports custom weekends
Simpler	More flexible
Suitable for standard offices	Suitable for custom work schedules
Optional holidays	Optional holidays

19. WORKDAY Function

The WORKDAY function returns a future or past working date after adding or subtracting a specified number of working days.

It automatically excludes Saturdays and Sundays and can optionally exclude holidays.

Purpose

Used to calculate business deadlines.

Key Difference

NETWORKDAYS answers:

How many working days are between two dates?

WORKDAY answers:

What will the date be after a specified number of working days?

Syntax

=WORKDAY(start_date,days,[holidays])

Syntax Explanation

Argument	Meaning
start_date	Starting date
days	Number of working days to add or subtract
[holidays]	Optional holiday dates to exclude

Basic Example

Suppose:

Start Date = 06-Jul-2026

You need the date after 5 working days.

Formula:

=WORKDAY(DATE(2026,7,6),5)

Result:
13-Jul-2026
Why not 10 July?
Because the start date itself is not counted as Day 1. Excel moves forward five working days:

- 07 July → Day 1
- 08 July → Day 2
- 09 July → Day 3
- 10 July → Day 4
- 13 July → Day 5

Saturday and Sunday are skipped.

Business Example Using Our Dataset

Suppose the Start Date is in D2, and company policy requires resolution within 5 working days.
Formula:
=WORKDAY(D2,5)
This calculates the expected deadline.

WORKDAY with Holidays

Suppose company holidays are stored in:
L2:L10
Formula:
=WORKDAY(D2,5,L2:L10)
Excel skips:

- Saturday
- Sunday
- All holidays listed in L2:L10

Calculating a Previous Working Date

WORKDAY also supports negative numbers.
Formula:
=WORKDAY(D2,-5)
This returns the date **5 working days before** the date in D2.

Business Uses of WORKDAY

- Project deadlines
- Invoice due dates
- Delivery dates
- Payment processing dates
- SLA deadlines
- Recruitment timelines
- Ticket resolution deadlines

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20. WORKDAY.INTL Function

The WORKDAY.INTL function calculates a future or past working date while allowing custom weekend patterns.

Purpose

Used when the organization doesn't follow the standard Saturday–Sunday weekend.

Syntax

=WORKDAY.INTL(start_date,days,[weekend],[holidays])

Syntax Explanation

Argument	Meaning
start_date	Starting date
days	Number of working days to add or subtract
[weekend]	Defines custom weekend days
[holidays]	Optional holiday range

Example: Friday and Saturday Weekend

=WORKDAY.INTL(D2,5,7)

Here:

- Start from D2
- Add 5 working days
- Treat Friday and Saturday as weekends

Example with Holidays

=WORKDAY.INTL(D2,10,1,L2:L10)

This calculates the date after 10 working days while:

- Excluding Saturday and Sunday.
- Excluding holidays in L2:L10.

WORKDAY vs WORKDAY.INTL

WORKDAY	WORKDAY.INTL
Saturday and Sunday are weekends	Supports custom weekend patterns
Simpler Syntax	More flexible
Standard office schedule	Custom business schedule

NETWORKDAYS vs WORKDAY

This is a frequently asked conceptual interview question.

NETWORKDAYS	WORKDAY
Returns a number	Returns a date
Counts working days between two dates	Calculates a date after/before N working days
Requires start and end dates	Requires start date and number of days
Used for duration analysis	Used for deadline calculation

Example

=NETWORKDAYS(D2,F2)

Question answered: How many working days did the ticket take?

=WORKDAY(D2,5)

Question answered: What is the deadline after 5 working days?

Real Business Scenario 1: Calculate Actual Working Days Taken

Dataset:

- D2 = Start Date
- F2 = End Date

Formula:

=NETWORKDAYS(D2,F2)

Possible result:

7

Interpretation:

The ticket took 7 working days from start to completion, including both boundary dates when they are workdays.

Real Business Scenario 2: Calculate Expected Deadline

Company policy:

Every ticket should be resolved within 5 working days.

Formula:

=WORKDAY(D2,5)

This returns the expected resolution deadline.

Real Business Scenario 3: Check SLA Compliance

Suppose:

- D2 = Start Date
- F2 = Actual End Date
- SLA = 5 working days

Formula:

=IF(F2<=WORKDAY(D2,5),"SLA Met","SLA Breached")

This is a highly practical combination of:

- IF
- WORKDAY

Real Business Scenario 4: Calculate SLA with Holidays

Suppose holidays are stored in L2:L10.
Formula:
=IF(F2<=WORKDAY(D2,5,L2:L10),"SLA Met","SLA Breached")
Now the calculation correctly excludes weekends and company holidays.

Real Business Scenario 5: Measure Delay in Working Days

Suppose:

- H2 = Due Date
- F2 = Actual End Date

A basic formula would be:
=IF(F2>H2,NETWORKDAYS(H2+1,F2),0)
Why H2+1?
Because NETWORKDAYS includes both start and end dates. If the ticket is completed one working day after the deadline, starting the count from H2+1 prevents the due date itself from being incorrectly counted as a delay day.
This detail matters in actual business reporting.

Important Interview Questions from This Module

1. What is the difference between DAYS and NETWORKDAYS?
DAYS counts all calendar days, whereas NETWORKDAYS excludes weekends and can optionally exclude holidays.
2. What is the difference between NETWORKDAYS and WORKDAY?
NETWORKDAYS returns the number of working days between two dates. WORKDAY returns a future or past date after adding or subtracting a specified number of working days.
3. When would you use .INTL versions?
When the organization has a custom weekend structure instead of the standard Saturday–Sunday weekend.
4. Does NETWORKDAYS count the start and end dates?
Yes, if they are valid working days.
5. Does WORKDAY count the starting date as Day 1?
No. It begins counting working days after the start date for positive day offsets.

Module 6 Summary

Function	Purpose	Syntax
NETWORKDAYS	Counts working days between two dates	=NETWORKDAYS(start_date,end_date,[holidays])
NETWORKDAYS.INTL	Counts working days with custom weekends	=NETWORKDAYS.INTL(start_date,end_date,[weekend],[holidays])
WORKDAY	Calculates a date after/before N working days	=WORKDAY(start_date,days,[holidays])
WORKDAY.INTL	Calculates a working date with custom weekends	=WORKDAY.INTL(start_date,days,[weekend],[holidays])

Core Formulas

=NETWORKDAYS(D2,F2)
=NETWORKDAYS(D2,F2,L2:L10)
=NETWORKDAYS.INTL(D2,F2,7)
=WORKDAY(D2,5)
=WORKDAY(D2,5,L2:L10)
=WORKDAY.INTL(D2,5,7)

Most Important Business Formula

=IF(F2<=WORKDAY(D2,5),"SLA Met","SLA Breached")

Module 7: Month-Based Date Calculations & Date Formatting

In the previous module, we covered working-day calculations using:

- NETWORKDAYS
- NETWORKDAYS.INTL
- WORKDAY
- WORKDAY.INTL

Now we'll move to functions used for **month-based calculations and professional date formatting**:

- EDATE
- EOMONTH
- TEXT

These functions are particularly useful in:

- Loan maturity calculations
- Subscription renewals
- Contract expiry dates
- EMI schedules
- Monthly financial reporting
- Billing cycles
- Month-end closing
- Dashboard labels
- Dynamic report generation

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21. EDATE Function

The EDATE function returns a date that is a specified number of **months before or after a given start date**. It is designed specifically for month-based date calculations.

Purpose

Use EDATE when you need to move forward or backward by a fixed number of months.
For example:

- 3 months after joining date
- 6 months after contract start
- 12 months after subscription date
- 2 months before a renewal date

Syntax

=EDATE(start_date,months)

Syntax Explanation

Argument	Meaning
start_date	The original date from which calculation begins
months	Number of months to add or subtract

Understanding the months Argument

- Positive number → moves into the future.
- Negative number → moves into the past.

- Zero → returns the same date.

Basic Example: Add 3 Months

Suppose:
Start Date = 15-Jan-2026
Formula:
=EDATE(DATE(2026,1,15),3)
Result:
15-Apr-2026

Example: Subtract 3 Months

Formula:
=EDATE(DATE(2026,7,15),-3)
Result:
15-Apr-2026
The negative value moves backward in time.

Business Example: Employee Probation Completion Date

Suppose:

- I2 = Employee Joining Date
- Probation Period = 6 months

Formula:
=EDATE(I2,6)
If:
Joining Date = 10-Jan-2026
Result:
10-Jul-2026
This calculates the employee's probation completion date.

Business Example: Annual Subscription Renewal

Suppose a customer's subscription begins on:
15-Mar-2026
The subscription is valid for 12 months.
Formula:
=EDATE(A2,12)
Result:
15-Mar-2027

Business Example: Contract Expiry

Suppose:

Start Date	Contract Duration
01-Apr-2026	18 months

Formula:
=EDATE(A2,B2)
Result:
01-Oct-2027
This is more reliable than manually adding a fixed number of days because months have different lengths.

Why Not Simply Add 30 Days?

Suppose you want a date exactly one month after:
31-Jan-2026
Using:
=A2+30
does **not** necessarily mean one calendar month later.
The correct approach is:
=EDATE(A2,1)
Result:
28-Feb-2026
because February 2026 has 28 days.
Important Business Rule
One month is not always 30 days.
For monthly contracts, subscriptions, EMIs, and renewals, use EDATE, not a fixed number of days.

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22. EOMONTH Function

The EOMONTH function returns the **last date of a month**, either for the current month or a specified number of months before or after a given date.
EOMONTH stands for:
End Of Month

Purpose

Used when the exact last date of a month is required.
This is particularly important in:

- Financial reporting
- Accounting
- Billing cycles
- Payroll
- Monthly dashboards
- Loan calculations
- Month-end closing

Syntax

=EOMONTH(start_date,months)

Syntax Explanation

Argument	Meaning
start_date	Reference date
months	Number of months before or after the start date

Example: Current Month-End

Suppose:
Start Date = 10-Jul-2026
Formula:
=EOMONTH(DATE(2026,7,10),0)
Result:
31-Jul-2026
The 0 means:
Return the last date of the same month.

Example: Next Month-End

Formula:
=EOMONTH(DATE(2026,7,10),1)
Result:
31-Aug-2026

Example: Previous Month-End

Formula:
=EOMONTH(DATE(2026,7,10),-1)
Result:
30-Jun-2026

Business Example: Financial Month-End Reporting

Suppose a transaction date is stored in D2.
Formula:
=EOMONTH(D2,0)
This assigns the transaction to its corresponding month-end date.
For example:

Transaction Date	Month-End Date
05-Jan-2026	31-Jan-2026
17-Feb-2026	28-Feb-2026
22-Mar-2026	31-Mar-2026

This is useful for monthly financial grouping and reporting.

Business Example: Calculate First Day of the Month

There is no need for a separate STARTOFMONTH function in regular Excel formulas.
You can calculate the first day of the current month using:
=EOMONTH(D2,-1)+1

How It Works

1. EOMONTH(D2,-1) returns the last day of the previous month.
2. Adding 1 moves to the first day of the current month.

Example

If:
D2 = 15-Jul-2026
Then:
=EOMONTH(D2,-1)+1
Result:
01-Jul-2026

Business Example: First Day of Next Month

Formula:
=EOMONTH(D2,0)+1
If:
D2 = 15-Jul-2026
Result:
01-Aug-2026

EDATE vs EOMONTH

This is an important conceptual and interview question.

EDATE	EOMONTH
Moves a date by a specified number of months	Returns the final date of a specified month
Generally preserves the day where possible	Always returns month-end
Useful for renewals and contracts	Useful for financial reporting
=EDATE(A2,3)	=EOMONTH(A2,3)

Example

Suppose:
A2 = 15-Jan-2026
Then:
=EDATE(A2,3)
returns:
15-Apr-2026
Whereas:
=EOMONTH(A2,3)
returns:
30-Apr-2026

The difference is fundamental.

23. TEXT Function for Date & Time Formatting

You already covered TEXT during the Text Functions lecture, so there is no need to reteach the entire function from scratch. However, it should still be included here because it is extremely important for **date and time formatting**.

The TEXT function converts a numeric value, date, or time into formatted text according to a specified format pattern.

Syntax

=TEXT(value,format_text)

Syntax Explanation

Argument	Meaning
value	Number, date, or time to format
format_text	Desired output format enclosed in quotation marks

Common Date Formats with TEXT

Suppose D2 contains:
10-Jul-2026

Formula	Result
=TEXT(D2,"dd")	10
=TEXT(D2,"ddd")	Fri
=TEXT(D2,"dddd")	Friday
=TEXT(D2,"mm")	07
=TEXT(D2,"mmm")	Jul
=TEXT(D2,"mmmm")	July
=TEXT(D2,"yy")	26
=TEXT(D2,"yyyy")	2026
=TEXT(D2,"dd-mmm-yyyy")	10-Jul-2026
=TEXT(D2,"mmmm yyyy")	July 2026

Common Time Formats with TEXT

Suppose E2 contains:
14:35:20

Formula	Result
=TEXT(E2,"hh:mm")	14:35
=TEXT(E2,"hh:mm:ss")	14:35:20
=TEXT(E2,"hh:mm AM/PM")	02:35 PM

Business Example: Create Month-Year Labels

Formula:
=TEXT(D2,"mmm-yyyy")
Result:
Jul-2026
This is useful for:

- Dashboard labels
- Monthly reporting
- Chart axes
- Summary reports

Important Analytics Warning

The result of TEXT is **text, not a genuine date**.
This matters because if you sort month names such as:
Apr-2026
Aug-2026
Dec-2026
Feb-2026
Excel may sort them alphabetically rather than chronologically, depending on how the values are used. For analytical models, it is often better to retain the original date column and use formatted text only for display.

Business Example: Dynamic Report Title

Formula:
="Ticket Performance Report - "&TEXT(TODAY(),"mmmm yyyy")
Possible result:
Ticket Performance Report - July 2026
This title updates automatically as the current month changes.

Business Example: Combine Employee Name with Joining Date

Suppose:

- B2 = Employee Name
- I2 = Joining Date

Formula:
=B2&" joined on "&TEXT(I2,"dd-mmm-yyyy")
Result:
Rahul Singh joined on 15-Mar-2021
Without TEXT, concatenating a date may expose Excel's underlying serial number rather than the desired formatted date.

Business Example: Month Name from Start Date

=TEXT(D2,"mmmm")
Result:

July
For an abbreviated month:
=TEXT(D2,"mmm")
Result:
Jul

Business Example: Day Name from Date

=TEXT(D2,"dddd")
Result:
Friday
For the abbreviated version:
=TEXT(D2,"ddd")
Result:
Fri

EOMONTH + TODAY: Dynamic Reporting Periods

These functions become especially powerful when combined.
Last Day of Current Month
=EOMONTH(TODAY(),0)
Last Day of Previous Month
=EOMONTH(TODAY(),-1)
Last Day of Next Month
=EOMONTH(TODAY(),1)
First Day of Current Month
=EOMONTH(TODAY(),-1)+1
First Day of Next Month
=EOMONTH(TODAY(),0)+1
These formulas are highly useful for dynamic dashboards and monthly reports.

Real Business Scenario 1: Employee Probation Status

Suppose:

- I2 = Joining Date
- Probation duration = 6 months

Formula:
=IF(TODAY()>=EDATE(I2,6),"Probation Completed","Under Probation")
Logic

- Calculate the date six months after joining.
- Compare it with today's date.
- If today is on or after that date → Probation Completed.
- Otherwise → Under Probation.

Real Business Scenario 2: Subscription Renewal Status

Suppose:

- A2 = Subscription Start Date
- Subscription duration = 12 months

Formula:

=IF(TODAY()>EDATE(A2,12),"Expired","Active")

For a more precise status that treats the exact expiry date as expired:

=IF(TODAY()>=EDATE(A2,12),"Expired","Active")

The correct operator depends on the business rule.

Real Business Scenario 3: Month-End Ticket Classification

Suppose the Start Date is in D2.

Formula:

=IF(D2=EOMONTH(D2,0),"Opened on Month End","Regular Date")

This checks whether a ticket was opened on the final calendar day of its month.

Real Business Scenario 4: Dynamic Reporting Month

=TEXT(TODAY(),"mmmm yyyy")

Possible result:

July 2026

Useful for dashboard headings and automated monthly reports.

Real Business Scenario 5: Calculate Next Review Date

Suppose an employee review occurs every 6 months from their joining date.

Formula:

=EDATE(I2,6)

For annual reviews:

=EDATE(I2,12)

However, for employees with multiple years of tenure, these formulas only calculate the **first** review date after joining. Calculating the **next upcoming recurring review date** requires more advanced logic.

Important Interview Questions from This Module

1. What is the difference between EDATE and EOMONTH?

EDATE shifts a date by a specified number of months while generally preserving the day of the month where possible. EOMONTH always returns the last date of the target month.

2. Why shouldn't you add 30 days to calculate one month later?

Because calendar months have different lengths. Use EDATE for month-based calculations.

3. How do you calculate the last day of the current month?

=EOMONTH(TODAY(),0)

4. How do you calculate the first day of the current month?

=EOMONTH(TODAY(),-1)+1

5. What is an important limitation of the TEXT function?

It converts the value to text. The result is no longer a genuine numeric date or time value for arithmetic Purposes.

Module 7 Summary

Function	Purpose	Syntax
EDATE	Moves a date forward or backward by months	=EDATE(start_date,months)
EOMONTH	Returns the last date of a target month	=EOMONTH(start_date,months)
TEXT	Converts dates and times into formatted text	=TEXT(value,format_text)

Core Formulas

=EDATE(I2,6)
=EOMONTH(D2,0)
=EOMONTH(D2,-1)+1
=TEXT(D2,"dd-mmm-yyyy")
=TEXT(D2,"mmmm yyyy")

Important Business Formulas

Employee Probation Status:
=IF(TODAY()>=EDATE(I2,6),"Probation Completed","Under Probation")
Dynamic Month-End:
=EOMONTH(TODAY(),0)
Dynamic Report Title:
="Ticket Performance Report - "&TEXT(TODAY(),"mmmm yyyy")
Month-Year Label:
=TEXT(D2,"mmm-yyyy")